

Text, Screenshots, or Both?

Representation Value and Task Routing in Web Agents

Anonymous ACL submission

Abstract

A web agent can read a page as text, look at a screenshot, or do both, and deployments usually fix one choice for every task. We measure that choice across three sites of VisualWebArena and WebArena with four vision-language backbones, using text-only variants as controls and rerunning identical conditions to see which differences survive. Three findings separate questions that are usually merged. First, no observation choice is best everywhere, and the screenshot has a real, reproducible value that is predictable in advance on one site: a zero-cost rule over the task text picks out classifieds tasks on which adding the screenshot alone raises success by 25–30 points for the two strongest backbones (11 for a 4B model), and the gap reappears on a full rerun. Second, which representation suits which task is a genuine property of the task, beyond its difficulty, yet a single run measures it with reliability of only 0.31–0.50. Third, routers that choose a representation per task, with their operating point fixed on training data, do not convert this signal into a gain over the best fixed choices and their random mixtures. We describe the rerun-controlled protocol that keeps these three quantities apart.

1 Introduction

An agent that operates a website has to perceive it, and the two standard ways are complementary in principle: the page’s text (an accessibility tree) is precise and cheap, a screenshot shows what the text omits. A third option fuses them, overlaying numbered marks on the screenshot and listing the marked elements as text (Yang et al., 2023; Koh et al., 2024). Practitioners typically pick one of the three for a whole deployment. The question this paper asks is the one that choice raises: *is the choice worth making per task, and can it be made?*

Answering it requires keeping apart three quantities that a single leaderboard number merges:

1. **Representation value:** does the screenshot change success, holding everything else fixed?
2. **Preference reliability:** when one representation beats another on a task, is that a stable property of the task, or a property of one run?
3. **Routing gain:** can a policy that chooses per task, using only what is known before it runs, beat the fixed choices?

A large hindsight ceiling — pick, for each task, whichever representation happened to succeed — answers none of them, because it rewards run-to-run luck as much as structure. We therefore rerun identical conditions and read every comparison against what a rerun alone changes.

Findings. (i) No choice wins everywhere: the fused mode leads on classifieds, text leads on WebArena reddit, and the order reverses across backbones (§3). (ii) The screenshot has a specific value that can be isolated and predicted: with the text payload and prompt held fixed, adding the screenshot raises success on the classifieds tasks a zero-cost intent rule flags by 25–30 points for the two strongest backbones (11 for a 4B model), against 5–10 on the other tasks; the effect reproduces on a full rerun and is largest for the strongest backbone. On reddit the same rule finds nothing (§4). (iii) Per-task preferences between representations exist beyond task difficulty — a no-interaction null conditioned on difficulty is rejected in every fully rerun cell — but a single run measures them with reliability 0.31–0.50; two to three runs per task are needed to pass 0.5 (§5). (iv) Routers that fix their operating point on training data do not beat the fixed representations and their random mixtures; the room that exists above that frontier comes from task difficulty, not from representation fit, and is not captured without looking at test outcomes (§6).

Contributions. A rerun-controlled measurement of text, screenshot and fused observation across

three sites and four backbones, with text-only ablations that isolate the screenshot; a matched estimate of the screenshot’s value and an ex-ante rule that predicts it on one site; a test of whether per-task representation preferences are real, with a decision study for how many runs make them learnable; and a deployable-policy comparison against fixed choices *and their random mixtures*, which is the comparison a routing claim has to win (§7).

2 Setup

Sites and tasks. We use three sites of VisualWebArena (Koh et al., 2024) — classifieds (224 scored tasks), reddit (203) and shopping (432) — and the reddit site of WebArena (Zhou et al., 2024) (104 tasks, those all observation modes ran). VWA-reddit and WA-reddit are different task sets on the same application, so they are not two independent benchmarks. Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration (Appendix A).

Backbones. B0 = Qwen3-VL-235B (API-served), B1 = Qwen3-VL-4B and B2 = Gemma-3-4B (both served locally), and B5 = GPT-5.6 (API, classifieds only). A *cell* is one site $\times$ backbone pair; the eight cells with every observation mode are VWA-classifieds and VWA-reddit $\times$ {B0, B1, B2} and WA-reddit $\times$ {B0, B1}. B5-classifieds and the two shopping cells extend them (§4, §6) and are never pooled with the eight.

One plain agent. One model call per step emits a thought, an action and its arguments; the prompt holds the instruction, the mode’s system prompt, the last eight steps and the current observation. There is no planner, memory, reflection or retry policy, so a difference between conditions is a difference of observation, not of scaffold. Episodes stop at 30 steps.

Observation modes. The deployment question has three answers: **DOM** (the page’s accessibility tree as text), **Vision** (the raw screenshot, with an empty text payload) and **SoM** (a set-of-marks screenshot plus the list of marked elements as text; Yang et al., 2023). Three further modes carry no screenshot and serve as controls; the one used throughout is **P-SoM**, which is SoM with the screenshot removed — same marks, same prompt — so SoM–P-SoM isolates the image (Appendix B lists all six). “Text” modes withhold the *page*

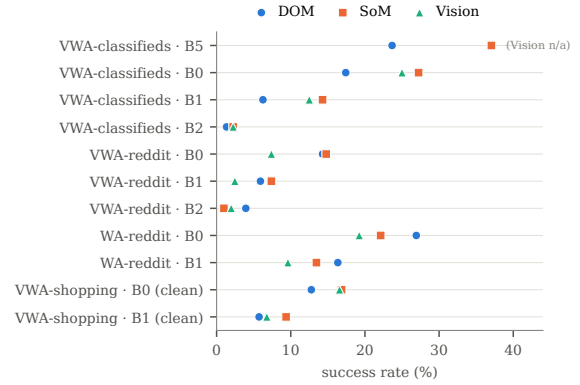


Figure 1: Success rate of the three deployment choices per cell. Shopping cells drop the tasks touched by a known harness defect in any mode (§2, Appendix D); B5’s Vision run is excluded for a coordinate-contract defect. Source: routing_three_arm, routing_extension_cells.

screenshot only: a reference image that is part of the task (“find this item”) is given in every mode.

Reruns. We reran identical conditions, 24 registered same-condition pairs in all; in three cells — VWA-classifieds B0, VWA-reddit B0 and WA-reddit B1 — every mode was run twice. A rerun uses the same tasks, code and site reset, so what it changes is what one run of the agent does not determine.

Scoring and cost. The evaluators are binary. Cost is the per-episode billed cost: an API invoice for B0 and B5, a token-priced electricity estimate for B1 and B2. The two kinds are ordered within a cell, never pooled across; §6 repeats the routing analysis under GPU-time and wall-clock cost.

3 No observation choice is best everywhere

Figure 1 shows the three deployment choices side by side. On classifieds the fused mode is the best of the three for B0, B1 and GPT-5.6 (B2 solves too few tasks to order the modes), and the margin over text is not small: SoM–DOM is +9.82 pp [+3.57, +16.07] for B0 and +8.04 pp [+3.57, +12.95] for B1 (paired task bootstrap). On WebArena reddit the order flips and text leads (SoM–DOM –4.81 pp for B0 and –2.88 pp for B1, both intervals crossing zero), and on VWA-reddit the weakest backbone does significantly worse with the fused mode (–2.96 pp [–5.91, –0.49]). On shopping the fused mode is best or tied with Vision (Figure 1).

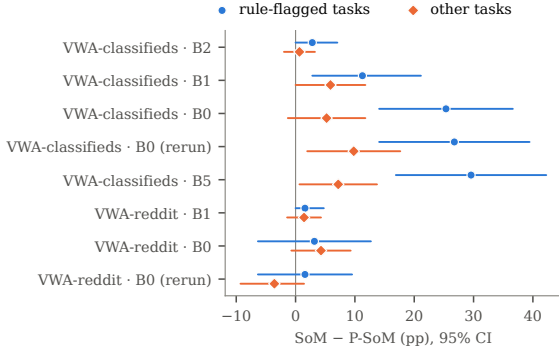


Figure 2: Value of the screenshot alone (SoM−P-SoM: same marks, same prompt) on the tasks a zero-cost intent rule flags, against the other tasks. Classifieds is out-of-sample for the rule. “Rerun” is a full repeat of both modes. VWA-reddit B2 is omitted: no mode solves a flagged task. Source: visual_intent_routing.

Two cautions keep this descriptive. Cells on the same site share their tasks, so the eight are not eight independent replications. And the screenshot-free modes are four, not one: a maximum over four text variants against a single screenshot mode would be a selection effect, so the comparison here takes one representative per deployment class, as practitioners do.

What this section establishes is only that the choice matters and depends on the workload. Whether it can be made per task is the subject of the rest of the paper.

4 The screenshot has a value that can be isolated and predicted

A matched contrast. Comparing Vision with DOM changes two things at once — the image and the text the agent receives. SoM and P-SoM differ only in the annotated screenshot, so their difference is the value of the image alone.

An ex-ante rule. A regular expression over the task intent flags tasks that ask about an image, a photo, a colour or a count of things shown, excluding tasks that already carry a reference image (Appendix C). It reads only the task configuration: no model call and no episode. It was written for reddit, so its classifieds hits were never inspected and classifieds is out-of-sample.

Result. On the 71 classifieds tasks it flags, the screenshot raises success by +25.35 pp [+14.08, +36.62] for B0, against +5.23 pp on the other 153 tasks (Figure 2). The effect reproduces on a full

rerun of both modes (+26.76 pp [+14.08, +39.44], from 23 against 4 successes), and it grows with capability: +2.82 pp for B2, +11.27 pp for B1, +25.35 pp for B0 and +29.58 pp [+16.90, +42.25] for GPT-5.6. On VWA-reddit the same rule flags 63 tasks and separates nothing: +3.17 pp on flagged tasks against +4.29 pp elsewhere for B0, +1.59 pp against −3.57 pp on its rerun. On shopping the flagged tasks carry at most five successes per mode, too few to read, and on WebArena the rule flags only 5 of 104 tasks.

Reading. Where the screenshot pays off on classifieds is predictable before anything runs, and it pays off more for stronger models; on reddit it is not. The screenshot’s value is therefore real and partly foreseeable, but site-specific: it is a statement about a kind of task, not a general law. It is also not a router. On these tasks the fused mode already contains the screenshot, so switching between DOM and Vision by this rule cannot beat simply using SoM; what remains open is whether *individual* tasks have stable preferences beyond such task types, which is the question of §5.

5 Per-task preferences are real, but one run cannot measure them

§4 found structure at the level of task *types*. Routing per task needs more: that an individual task prefers one representation over another, beyond being easy or hard, and that this preference is stable enough to learn from.

A null without task-by-mode structure. In the three cells where every mode was run twice, let $Y_{imr} \in \{0, 1\}$ be the outcome of task i under mode m in run r . Under

$$\text{logit } \Pr(Y_{imr} = 1) = a_i + b_{mr},$$

every task has its own difficulty and every (mode, run) column its own easiness, but there is no persistent task-by-mode interaction. The row and column sums are sufficient for this model, so conditional on them every 0/1 matrix with those margins is equally likely; we sample that null exactly with curveball swaps, fitting nothing. The statistic is the reproducible part of the interaction, $\hat{\sigma}_{\text{int}}^2 = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}})/2$ from the two-way task $\times$ mode ANOVA with two runs: the run-to-run covariance of each task’s mode profile.

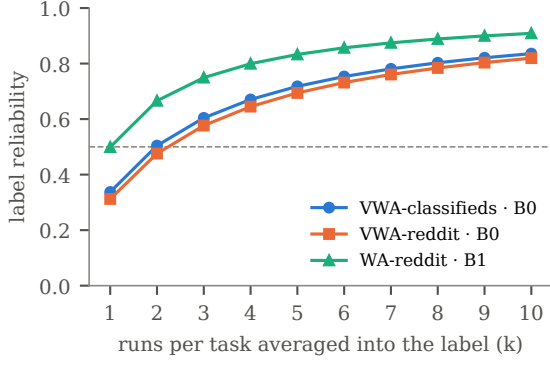


Figure 3: Decision study for the three deployment modes: reliability of a task-by-mode label averaged over k runs, $\sigma_{\text{int}}^2 / (\sigma_{\text{int}}^2 + \sigma_{\text{err}}^2 / k)$. A single run gives 0.31–0.50. Source: routing_three_arm.

Result. Over the three deployment modes the interaction exceeds the null in all three cells (classifieds B0 $\hat{\sigma}_{\text{int}}^2 = 0.033$ against a null 95th percentile of 0.015; reddit B0 0.018 against 0.009; WebArena-reddit B1 0.030 against 0.014; $p \leq 0.002$ each, Holm across cells). Over all six modes the same holds (Appendix E). Preferences between representations are therefore a property of tasks, not only of their difficulty.

But a single run measures them poorly. The same decomposition gives the reliability of a label built from k runs (Figure 3). From one run it is 0.34, 0.31 and 0.50 in the three cells; two to three runs per task are needed to pass 0.5. Over six modes it is lower still, 0.14–0.46, and up to six runs are needed. The preferences also persist where they can be observed directly: taking, for each task, the cheapest mode that succeeded on one run and scoring that choice on the rerun beats the fixed-mode frontier by 2.4–10.5 points on reddit B0 and WebArena-reddit B1. That choice uses the task’s own label, which no deployment has, so it measures persistence, not an attainable policy.

The routing target thus exists, and a training set built from single runs labels it with a reliability between a third and a half.

6 Deployable routers do not convert the signal

The comparison a router must win. A router that sends some tasks to an expensive mode and others to a cheap one trades success for cost. Beating every fixed mode on both axes is too strict, but beating the fixed modes one at a time is too

lenient: sending a random share of tasks to each of two modes is also a deployable policy. We therefore compare against the *frontier of the fixed modes and all their random mixtures* — the upper concave envelope of the fixed points in the (cost, success) plane — and report how far above it a router reaches.

Routers. Two constructions use 18 features available before a task runs: 14 intent-keyword indicators, the intent length, and three statistics of the first page. *Per-arm heads* fit one logistic success model per mode and send a task to the cheapest mode whose predicted success clears a threshold. *Triage* predicts whether any mode can solve the task, sends predicted-hopeless tasks to the cheapest mode and the rest to the best one. All predictions are out-of-fold (task-held-out five-fold).

Room above the frontier exists, and it is difficult. Sweeping each router’s threshold traces a curve. Over the three deployment modes, the triage curve pooled over the eight cells rises 1.69 points above the frontier at low budget, which a label-shuffle null that refits the whole curve does not produce ($p = 0.001$; Table 1). The per-arm heads reach 0.86 points ($p = 0.072$). Triage does not choose a representation for a task; it decides which tasks are not worth the expensive mode. The room that exists is the difficulty lever, not representation fit.

A deployable router does not capture it. A curve’s best point is chosen after seeing test outcomes. When the operating point is instead chosen on training folds only, the pooled gain is +0.23 points for per-arm heads and +0.59 for triage, and neither interval excludes zero (Table 1). Over all six modes the deployable per-arm router sits significantly *below* the frontier (−0.38; 95% upper bound −0.14). The pattern holds for the strongest backbone: for GPT-5.6 on classifieds, where the screenshot’s value was largest (§4), the deployable routers lose 1.6–2.5 points; on both shopping cells they stay within 1.2 points of the frontier.

Robustness. Holding tasks of the same template out together lowers the six-mode per-arm router’s mean deployable gain from −0.38 to −1.60 points, and fitting on one run and scoring on its rerun gives point estimates of both signs whose intervals all include zero (Appendix F). Repricing cost as GPU time or wall-clock time leaves the frontier result unchanged, but changes *which* single cell’s

cells	policy	curve max (p)	deployable [5%, 95%]
3 arms, 8 cells pooled	per-arm heads	+0.86 (0.072)	+0.23 [−2.20, +0.22]
	triage	+1.69 (0.001)	+0.59 [−1.53, +1.23]
6 arms, 8 cells pooled	per-arm heads	+0.67 (0.252)	−0.38 [−3.20, −0.14]
	triage	+1.24 (0.043)	−0.39 [−2.42, +0.81]
GPT-5.6, classifieds (5 arms)	per-arm heads	+0.45 (0.768)	−2.45 [−8.61, +4.24]
	triage	+3.86 (0.127)	−1.63 [−6.70, +5.74]
Shopping B0 (3 arms)	per-arm heads	+1.14 (0.062)	−0.18 [−2.45, +2.02]
	triage	+0.20 (0.496)	−0.87 [−2.63, +1.37]
Shopping B1 (6 arms)	per-arm heads	+1.11 (0.143)	+0.56 [−4.69, +0.88]
	triage	+1.63 (0.158)	+1.19 [−4.10, +1.57]

Table 1: Routing gain over the fixed arms and their random mixtures, in SR points. *Curve max*: the best point of an out-of-fold curve, chosen after seeing test outcomes, with its label-shuffle p (pooled rows: max over a normalised budget). *Deployable*: the operating point chosen on training folds only, task bootstrap. Sources: routing_three_arm, representation_routing_frontier, routing_gain_upper_bounds, routing_extension_cells.

triage curve passes a per-cell Holm correction, so no single-cell result is quoted here. One of the three-mode intervals (per-arm heads, pooled) has an upper end below its point estimate, a sign that the bootstrap of a policy with a selected threshold is biased; we read these intervals as magnitudes, not as tests.

7 A protocol for multi-representation claims

The three quantities of §1 came apart because each was measured against its own control. We suggest the same four steps for any claim that choosing among several agent configurations per task would pay.

1. **Report reruns per arm and per cell**, not one flip rate or one resolution threshold for a whole study: in our data a rerun changes between 0 and 14% of outcomes depending on arm and cell (Appendix E).
2. **Isolate the component of interest with a matched ablation** (here SoM–P-SoM, which removes only the screenshot) before attributing a difference to it.
3. **Test for per-task structure against a null that conditions on difficulty**, and use the decision study to say how many runs per task a training label needs. A single-run oracle is not a ceiling a router can reach.
4. **Score a deployable policy, operating point chosen on training data, against the fixed arms and their random mixtures**, never only against the best single arm or a hindsight curve.

Applied here, the protocol says the screenshot is worth paying for on a predictable class of classifieds tasks, that per-task representation preferences are real but need two to three runs per task to be learnable, and that with single-run labels and pre-flight features a per-task representation router does not yet beat choosing one representation, or a fixed mixture, for the whole workload.

8 Related work

[TODO: chunk 3: only citations already in paper.bib.]

Limitations

[TODO: chunk 3: required by ARR.]

References

- Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. 2024. [Visualwebarena: Evaluating multimodal agents on realistic visual web tasks](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. 2023. [Set-of-mark prompting unleashes extraordinary visual grounding in gpt-4v](#). *Preprint*, arXiv:2310.11441.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. [Webarena: A realistic web environment for building autonomous agents](#). In *International Conference on Learning Representations*.

A Task universes and exclusions

[TODO: chunk 3]

B The six observation modes

[TODO: chunk 3]

C The intent rule

[TODO: chunk 3]

D Harness defects and how cells were filtered

[TODO: chunk 3]

E Six-mode results

[TODO: chunk 3: six-mode null and D-study (task_mode_interaction_null), per-arm rerun discordance (noise_floor_inventory).]

F Template and rerun robustness

[TODO: chunk 3: routing_crossrun_template_validation and routing_three_arm template / $A \rightarrow B$ columns.]